

Proof-Carrying Budgets for Anonymous DHTs

Dual certificates and tiny verifiers for padding, delegation, and congestion contracts

February 2026

Abstract

Anonymous DHT designs increasingly resemble *compiled systems*: a lookup compiler chooses padding/deadlines, delegation lengths, retry rules, and congestion knobs, then an “accountant” reports a privacy budget. If budgets are meaningful, they should be verifiable by parties who do *not* trust the compiler.

We present a proof-carrying pattern for privacy budgets. A deployment publishes a small *certificate* consisting of: (i) a concrete plan (primal witness) and (ii) a dual witness that upper-bounds the privacy objective. A tiny verifier checks feasibility and the duality inequality, reproducing the declared bound without re-running large experiments.

This paper focuses on the *math interface*: dual certificates for TV/privacy objectives and how to package them into ABOM/OINL and release receipts. Supply-chain mechanics (receipts, transparency anchoring, many-testing) are treated as imported modules from other papers in this archive.

Scope. We isolate and formalize *Proof-Carrying Budgets for Anonymous DHTs* as a reusable contract surface in the anonymity / Anonymous-DHT series.

Imports. We treat (i) padding duals/transport certificates as owned by the published [[2026.01.29 - Mathematics: Optimal Poset-Feasible Padding: Knapsack, Transport, and Dual Certificates for TV Privacy]], (ii) stop-time normal forms as owned by [[2026.01.26 - Mathematics: Stop-Time Padding Addendum, Max-Mixture Separations and Tail-Sign Deadline Normal Forms]], and (iii) claim packaging as owned by ABOM/OINL and Release receipts.[?, ?]

Exports. A certificate interface (π, λ) and verifier checklist for TV-style budget claims, suitable for inclusion in ABOM/OINL and for anchoring in release receipts.[?, ?]

Artifact policy. This archive is source-only; supporting artifacts are available on request. See [?].

1 The proof-carrying template

Many “privacy budgets” in this series can be written as optimization problems: choose a mechanism π (a padding schedule, a randomized delegation rule, a congestion calendar) to minimize some cost subject to a privacy constraint, or dually maximize privacy subject to a cost envelope. When the budget is stated in *TV form* or another convex divergence, these problems admit duals whose feasibility can be checked locally.

Definition 1 (Certificate (informal)). A budget certificate is a tuple (π, λ) where π is a concrete mechanism instance (a plan) and λ is a dual witness such that: (i) π satisfies the stated feasibility constraints; (ii) λ is dual-feasible; and (iii) the dual objective evaluated at λ upper-bounds the privacy objective of π .

Lemma 1 (What the verifier checks). If the verifier can (a) re-evaluate the primal objective/constraints and (b) re-evaluate the dual objective/constraints from (π, λ) , then the verifier can reproduce the published privacy bound with no access to private traces.

2 A minimal verifier interface

The verifier interface is intentionally small; it should fit in a few hundred lines.

Algorithm 1 Toy verifier interface for a budget certificate

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1: procedure VERIFY( $\mathcal{P}, \mathcal{D}, \pi, \lambda, B$ )
2:   check  $\pi \in \mathcal{P}$  (schema + feasibility)
3:   check  $\lambda \in \mathcal{D}$  (schema + dual feasibility)
4:    $p \leftarrow \text{PrimalValue}(\pi)$ 
5:    $d \leftarrow \text{DualValue}(\lambda)$ 
6:   check  $p \leq d$  ▷ weak duality / certificate inequality
7:   check  $d \leq B$  ▷ published bound
8:   return ok
9: end procedure

```

What counts as “primal” here? For padding, π can be a discrete distribution over stop times (or a mixture over deadline normal forms). For committee contact privacy, π can be a randomized contact schedule (with public randomness commitments). For congestion, π can be the compiler knob manifest plus the accountant transcript.

3 Composing certificates with labels and receipts

Proof objects must be publishable. We recommend the following layering (owned by other papers):

- **ABOM/OINL:** ABOM commits to (π, λ) by digest and states what claim the certificate supports; OINL summarizes the resulting budget.[?]
- **Release receipts:** a release receipt binds the ABOM (and its digests) to a particular tagged release, with policy-as-code gates and optional transparency anchoring.[?]
- **Specialized certificates:** congestion budgets add log inclusion/consistency proofs and monitoring receipts (W-Congestion-EQ).[?]

4 Limitations

A dual certificate validates arithmetic and consistency, not ground truth. If the underlying model misses a side channel, a perfect certificate can still describe the wrong world. We therefore treat proof-carrying budgets as one layer in a larger publication pipeline that also includes state-dependent threat windows and closed-view auditing in later papers.

5 Takeaways

- **Proof-carrying budget pattern:** publish a concrete plan π and a dual witness λ so third parties can re-check the stated TV/privacy bound by weak duality.
- **Keep verifiers tiny:** the verifier only needs schema checks and objective evaluation; it should not need private traces or large simulations.
- **Packaging is layered:** ABOM/OINL commits to (π, λ) by digest and states the claim; release receipts bind that commitment to a tagged release; specialized series (e.g., W-Congestion-EQ) add log receipts and monitoring proofs.
- **Limits:** certificates validate arithmetic and consistency, not whether the model omitted a side channel—pair with threat-window analysis and auditing.

References

- [1] Anonymous. Artifact and Disclosure Policy for the One-True-Archive (Source-Only Public Release). Series synthesis note (Synthesis 6), 2026. (This archive.)
- [2] Anonymous. ABOM and OINL for Anonymity Claims: Proof-Carrying Odds-Inflation Budgets Without Publishing Full Artifacts. Anonymity series Paper C, 2026. (This archive.)
- [3] Anonymous. Privacy as a Release Property: Proof-Carrying Anonymity Receipts for Anonymous DHTs. Release and destination Paper A, 2026. (This archive.)
- [4] Anonymous. W-Congestion-EQ: Proof-Carrying Congestion Budgets with Certificates and Transparency Logs for Anonymous Overlays. Congestion series Paper 3, 2026. (This archive.)